Module/Pathway/Taxon Review: L-Arginine Biosynthesis via Acetylated Ornithine in *Pseudomonas putida* KT2440

Target pathway: KEGG ppu00220 "Arginine biosynthesis" (acetylated-ornithine microbial route) **Target taxon:** *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module area:** amino_acid_metabolism

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1. Executive Summary

The acetylated-ornithine route to L-arginine is **complete and satisfiable** in *Pseudomonas putida* KT2440. All eight canonical steps of the module — from L-glutamate acetylation through argininosuccinate lysis to L-arginine — are encoded by identifiable genes in the KT2440 genome, and the strain is a documented prototroph, providing physiological confirmation that the pathway is functional. The module should therefore be marked **COVERED**.

However, the 31-gene candidate list supplied by local metadata is a substantial over-collection: only roughly **10–11 of the 31 candidates are true de novo arginine-biosynthesis pathway members**. The remainder are over-propagated or out-of-boundary assignments that co-map to KEGG map00220 for reasons of shared nodes or reaction chemistry: the entire catabolic arginine deiminase (ADI) operon (*arcABC*), carbamoyl-phosphate synthase (*carAB*, a shared precursor node explicitly outside the module boundary), urease (*ureABC*), five glutamine-synthetase / polyamine-ligase paralogs, and several glutamate/glutamine-handling enzymes (*gdhA*, *gdhB*, *alaA*, *ansB*). These should be flagged as **not_in_module** during curation.

Three curation-critical nuances emerged. First, the **K00611 paralog ambiguity** (ornithine carbamoyltransferase) is resolved by regulatory evidence from *Pseudomonas*: PP_1079 (*argF*) is the anabolic, arginine-repressed biosynthetic OTCase (module step 6), while PP_1000 (*arcB*) is the catabolic OTCase of the ADI operon and is **out of module**. Second, KT2440 has **no dedicated E. coli-type ArgD** (K00818) for step 4; the acetylornithine aminotransferase activity is supplied by promiscuous class-III AST-pathway transaminases. Third — and important for curation completeness — the candidate metadata **omits PP_4481 (argD/astC, K00840)**, a second dual-function transaminase relevant to step 4 that should be added to the module gene set. Redundancy is a recurring theme: dedicated ArgA plus bifunctional ArgJ at step 1, hydrolytic ArgE plus ArgJ transacetylase at step 5, and two ArgC paralogs at step 3.

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2. Target-Organism Pathway Definition

What is included

The module covers **de novo microbial L-arginine biosynthesis from L-glutamate through N-acetylated intermediates and L-ornithine**. The biochemical scope is the eight-reaction sequence:

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L-glutamate
   │ (1) ArgA / ArgJ          N-acetylglutamate synthase
N-acetyl-L-glutamate
   │ (2) ArgB                 acetylglutamate kinase
N-acetyl-L-glutamyl-5-phosphate
   │ (3) ArgC                 N-acetyl-γ-glutamyl-phosphate reductase
N-acetyl-L-glutamate-5-semialdehyde
   │ (4) ArgD / AruC          acetylornithine aminotransferase
N-acetyl-L-ornithine
   │ (5) ArgE / ArgJ          ornithine release (hydrolysis or transacetylation)
L-ornithine
   │ (6) ArgF                 ornithine carbamoyltransferase (anabolic)
L-citrulline
   │ (7) ArgG                 argininosuccinate synthase
argininosuccinate
   │ (8) ArgH                 argininosuccinate lyase
L-arginine
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P. putida encodes both the **cyclic (ArgJ-dependent) implementation** and elements of the **linear (ArgA/ArgE) implementation** — the two are not mutually exclusive here, and the genome carries redundant machinery for the acetyl-management steps (1 and 5). This dual capacity is well-documented across *Pseudomonas* and consistent with the KEGG modules M00028 (glutamate ⇒ ornithine) and M00844 (ornithine ⇒ arginine).

What must be kept separate

Several neighboring processes co-locate on the broad KEGG overview map00220 but must **not** be conflated with de novo biosynthesis for this module:

- **Catabolic arginine deiminase (ADI) pathway** (*arcABC*): runs arginine → ornithine + carbamoyl-P → CO_2 + ATP, the reverse physiological direction.
- **Arginine succinyltransferase (AST) catabolic pathway** (*aru* genes): the major aerobic arginine/ornithine utilization route in *Pseudomonas*; it shares transaminase enzymes with biosynthesis but is a distinct pathway (ppu00330).
- **Carbamoyl-phosphate synthesis** (*carAB*): a shared substrate node with pyrimidine metabolism, explicitly **outside** the module boundary per the working scope.
- **Urea cycle / urease** (*ureABC*): urea amidohydrolase chemistry, unrelated to de novo arg biosynthesis.
- **Glutamate/glutamine precursor supply** (glutamine synthetase paralogs, glutamate dehydrogenases): upstream precursor metabolism, outside module boundary.

Alternate names / database definitions

- **KEGG:** ppu00220 "Arginine biosynthesis"; modules **M00028** (ornithine biosynthesis, glutamate ⇒ ornithine) and **M00844** (arginine biosynthesis, ornithine ⇒ arginine).
- **Route distinction:** the module intentionally excludes the succinylated-intermediate route, the N-acetylcitrulline route, and the **LysW-dependent** arginine/lysine route — these are distinct implementations not expanded here and not used by *P. putida* for this pathway.

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3. Expected Step Model

Step	Reaction	Expected family	KT2440 gene(s)	KO	Disposition
1	L-Glu → N-acetyl-L-Glu	ArgA (dedicated) **or** ArgJ (bifunctional)	*argA*/PP_5185; *argJ*/PP_1346	K14682; K00620	**covered** (redundant)
2	N-acetyl-L-Glu → N-acetyl-L-Glu-5-P	ArgB	*argB*/PP_5289	K00930	**covered**
3	N-acetyl-L-Glu-5-P → N-acetyl-L-Glu-5-semialdehyde	ArgC (type 1/2)	*argC1*/ PP_0432; *argC2*/ PP_3633	K00145	**covered** (2 paralogs)
4	semialdehyde → N-acetyl-L-ornithine	ArgD / promiscuous AST transaminase	*aruC*/PP_0372; + ***argD*/ PP_4481 (add)**	K00821; K00840	**covered** (via promiscuous enzymes)
5	N-acetyl-L-ornithine → L-ornithine	ArgE (hydrolytic) **or** ArgJ (transacetylase)	*argE*/PP_5186; *argJ*/PP_1346; PP_3571 (uncertain)	K01438; K00620	**covered** (redundant)
6	L-ornithine → L-citrulline	ArgF (anabolic OTCase)	*argF*/PP_1079	K00611	**covered**
7	L-citrulline → argininosuccinate	ArgG	*argG*/PP_1088	K01940	**covered**
8	argininosuccinate → L-arginine	ArgH	*argH*/PP_0184	K01755	**covered**

Every step is encoded. The step-ordering constraints in the module definition (1→2→3→4→5→6→7→8) are all satisfied.

4. Candidate Genes and Evidence

4.1 Core pathway genes (high confidence — module members)

Step 1 — N-acetylglutamate synthase. Two independent implementations are present. *argA*/**PP_5185** (UniProt **P0A0Z9**, K14682) is the dedicated ArgA/NAGS. This assignment is verified: a UniProtKB lookup of P0A0Z9 returns organism "*Pseudomonas putida* (strain KT2440)", gene *argA*/PP_5185, "Amino-acid acetyltransferase (NAGS)", and an organism_id:160488 + gene:argA search returns exactly this entry (F005). Comparative evidence from the close relative *P. aeruginosa* establishes that these organisms carry a classical AAK-GNAT NAGS: *"In many microorganisms, the first step of arginine biosynthesis is catalyzed by the classical N-acetylglutamate synthase (NAGS), an enzyme composed of N-terminal amino acid kinase (AAK) and C-terminal histone acetyltransferase (GNAT) domains"* (PMID: 22447897). The second implementation, *argJ*/**PP_1346** (UniProt **P59612**, K00620), is confirmed bifunctional (α/β cleaved chains; glutamate N-acetyltransferase EC 2.3.1.35 + amino-acid acetyltransferase EC 2.3.1.1), providing the acetyl-CoA-dependent initiating activity (F005). Cross-species mechanistic support that ArgJ operates at both step 1 and step 5 comes from *N. gonorrhoeae*, whose *argJ* complements both *E. coli argE* and *argA* mutations (PMID: 1339419).

Step 2 — Acetylglutamate kinase. *argB*/**PP_5289** (UniProt **P59300**, K00930), acetylglutamate kinase / NAGK (EC 2.7.2.8). Confirmed KT2440 entry (F005). Unambiguous, single-copy assignment.

Step 3 — N-acetyl-γ-glutamyl-phosphate reductase (AGPR). Two paralogs: *argC1*/**PP_0432** (Q88QQ6) and *argC2*/**PP_3633** (P59308), both K00145 (EC 1.2.1.38). This matches the module's "Type 1 / Type 2 ArgC" alternative-versions design. Both are annotated NAGSA dehydrogenases; either likely suffices for the step, so the step is robustly covered.

Step 4 — Acetylornithine aminotransferase. *No dedicated E. coli-type ArgD (K00818) exists in KT2440.* The activity is supplied by promiscuous class-III AST-pathway transaminases. *aruC*/**PP_0372** (Q88QW2, K00821, EC 2.6.1.11/2.6.1.13) is the only step-4 candidate in the 31-gene list, but its **primary local bucket is ppu00300** (lysine biosynthesis), signalling a cross-pathway/promiscuous assignment (it also carries N-succinyl-diaminopimelate aminotransferase activity, EC 2.6.1.17). In *P. aeruginosa* PAO1, *aruC* encodes the AST-pathway N2-succinylornithine 5-aminotransferase — the major aerobic arginine utilization route — and the same transaminase family supplies biosynthetic acetylornithine transamination: *"The arginine succinyltransferase*

(AST) pathway is the major arginine and ornithine utilization (aru) pathway under aerobic conditions in Pseudomonas aeruginosa" and *"the aruR, aruC, aruD, aruB, and aruE genes specify the ArgR regulatory protein, N2-succinylornithine 5-aminotransferase"* (PMID: 9393691) (F002).

Step 5 — Ornithine release. Two mechanisms are present. Hydrolytic: *argE*/**PP_5186** (Q88CJ5, K01438, acetylornithine deacetylase EC 3.5.1.16). Transacetylation: the bifunctional *argJ*/PP_1346 ornithine acetyltransferase (OAT) activity recycles the acetyl group. A third candidate, **PP_3571** (Q88GZ4, "Acetylornithine deacetylase"), is of **uncertain** status (see §5). The redundancy of ArgE + ArgJ means the step is robustly covered regardless of PP_3571's true function.

Step 6 — Ornithine carbamoyltransferase (anabolic). *argF*/**PP_1079** (Q88NX4, K00611, EC 2.1.3.3). The K00611 paralog ambiguity is resolved by regulatory evidence: in *P. aeruginosa* PAO1, the arginine-inducible regulator ArgR binds the control regions of the *car* and *argF* operons, and *argF* encodes the anabolic OTCase required for arginine biosynthesis: *"an arginine-inducible DNA-binding protein that interacts with the control regions for the car and argF operons, encoding carbamoylphosphate synthetase and anabolic ornithine carbamoyltransferase, respectively. Both enzymes are required for arginine biosynthesis"* (PMID: 9286980) and *"the arginine-repressible gene for anabolic ornithine carbamoyltransferase"* (PMID: 1538697). By orthology, PP_1079 is the biosynthetic OTCase, whereas PP_1000 (*arcB*) is catabolic (F004). This is reinforced by *P. syringae*, where *argF*/OCTase is negatively regulated by *argR* similarly to *P. aeruginosa* (PMID: 15150254).

Step 7 — Argininosuccinate synthase. *argG*/**PP_1088** (P59604, K01940, EC 6.3.4.5). Single-copy, unambiguous.

Step 8 — Argininosuccinate lyase. *argH*/**PP_0184** (P59618, K01755, EC 4.3.2.1). Confirmed KT2440 entry (F005). Single-copy, unambiguous; the terminal enzyme.

4.2 Metadata gap — gene to add

PP_4481 (*argD*/*astC*, UniProt P59319, K00840). A UniProt organism_id:160488 search returns **two** acetylornithine aminotransferases: Q88QW2 (*aruC*/PP_0372, in the candidate list) and **P59319 (argD/PP_4481, EC 2.6.1.11), which is NOT in the 31-gene candidate list** (F006). KEGG assigns PP_4481 to K00840 (*astC*, succinylornithine aminotransferase EC 2.6.1.81, pathway ppu00330, module M00879 AST pathway). Its *P. aeruginosa* ortholog PA0895 is the biochemically characterized AST N2-succinylornithine 5-aminotransferase (PMID: 9393691). Because both KT2440 transaminases are promiscuous dual catabolic/biosynthetic class-III enzymes, PP_4481 is a legitimate step-4 contributor and should be added to the module gene set for completeness.

5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 Out-of-boundary / over-propagated candidates (flag not_in_module)

Of the 31 candidates, approximately 20 are not de novo arginine-biosynthesis members (F003):

Category	Genes	KO	Why out of boundary
Catabolic ADI operon	*arcA*/PP_1001, *arcB*/PP_1000, *arcC*/PP_0999	K01478, K00611, K00926	Arginine deiminase pathway — opposite direction (arginine degradation → ATP)
Carbamoyl-P synthase	*carA*/PP_4724, *carB*/PP_4723	K01956, K01955	Shared pyrimidine node; explicitly outside module scope
Urease	*ureA/B/C*/PP_2843-45	K01430/ K01429/ K01428	Urea amidohydrolase; urea-cycle node of map00220, not biosynthesis
Glutamine synthetase / polyamine ligase paralogs	*glnA*/PP_5046, PP_3148, PP_4399, PP_4547, *puuA-I*/ PP_2178, *puuA-II*/PP_5299, *spuB*/PP_5183, *spuI*/PP_5184	K01915	Glutamate/glutamine and polyamine precursor supply — upstream of module
Glutamate handling	*gdhA*/PP_0675, *gdhB*/ PP_2080, *alaA*/PP_1872, *ansB*/PP_2453	K00262, K15371, K14260, K05597	Precursor / amino-acid interconversion, not de novo arg biosynthesis

The catabolic *arcABC* operon deserves emphasis: *arcB*/PP_1000 shares K00611 with the biosynthetic *argF* but is embedded in the ADI operon alongside *arcA* (arginine deiminase, K01478) and *arcC* (carbamate kinase, K00926) and runs toward citrulline degradation and ATP generation (F004). Transcriptomic data show KT2440 induces arginine-fermentation / ADI genes under anoxic and stress conditions (PMID: 23036925, PMID: 23537069), confirming these are physiologically catabolic rather than biosynthetic.

5.2 Genuine ambiguities

- **PP_3571 (second "acetylornithine deacetylase", Q88GZ4):** annotated as an ArgE-family deacetylase but of uncertain physiological role. With *argE*/PP_5186 and ArgJ already covering step 5, PP_3571's contribution is redundant/uncertain — mark **candidate_uncertain**. It may be a substrate-promiscuous amidohydrolase rather than the biosynthetic enzyme.
- **Step 4 broad EC/GO mappings:** both PP_0372 (K00821, EC 2.6.1.11/2.6.1.13/2.6.1.17) and PP_4481 (K00840, EC 2.6.1.81/2.6.1.11) carry broad, multi-pathway EC assignments. Neither is a "clean" dedicated ArgD; both are promiscuous AST-type class-III transaminases. This is a genuine lineage-specific alternative, not an over-annotation to discard — but it should be curated as "promiscuous enzyme fills step" rather than "dedicated ArgD present."
- **Step 1/5 ArgJ vs ArgA/ArgE redundancy:** whether the cyclic (ArgJ) or linear (ArgA+ArgE) route dominates in vivo is not resolved from sequence alone; both are encoded.

5.3 Likely annotation over-propagation

The primary over-propagation driver is KEGG map00220's inclusion of urea-cycle and catabolic nodes, plus broad K01915 (glutamine synthetase) mapping. The single most consequential curation error to avoid is treating *arcB*/PP_1000 as the biosynthetic OTCase; regulatory evidence firmly assigns that role to *argF*/PP_1079.

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6. Module and GO-Curation Recommendations

Overall module disposition: COVERED (satisfiable). All 8 steps are encoded and KT2440 is a documented prototroph (F001, F007).

Per-step curation marks:

Step	Mark	Notes
1	**covered**	argA/PP_5185 (dedicated) + argJ/PP_1346 (bifunctional)
2	**covered**	argB/PP_5289
3	**covered**	argC1/PP_0432 + argC2/PP_3633 (two paralogs)
4	**covered** (via promiscuous enzymes)	aruC/PP_0372 + **add argD/PP_4481**; no dedicated E. coli-type ArgD
5	**covered**	argE/PP_5186 + argJ OAT; PP_3571 = candidate_uncertain
6	**covered**	argF/PP_1079 (anabolic); arcB/PP_1000 = not_in_module
7	**covered**	argG/PP_1088
8	**covered**	argH/PP_0184

Actions: 1. **Add PP_4481 (argD/astC, K00840, UniProt P59319)** to the module gene set as a step-4 contributor. 2. **Flag ~20 candidates as not_in_module:** *arcABC*, *carAB*, *ureABC*, the eight K01915 GS/polyamine paralogs, *gdhA*, *gdhB*, *alaA*, *ansB*. 3. **Mark PP_3571 candidate_uncertain** (redundant/ambiguous step-5 deacetylase). 4. **Module boundaries are correct for this organism** — no module_needs_revision at the structural level. The generic scope (excluding carbamoyl-P, succinyl/LysW routes) fits KT2440 well. The only refinement is documenting that step 4 is served by promiscuous AST-type transaminases rather than a dedicated ArgD, which may warrant a GO annotation note (acetylornithine aminotransferase activity, GO:0003992, attributed to promiscuous class-III enzymes). 5. No new module document is required; a curation note on the ArgJ/ArgA + ArgE dual-implementation and the promiscuous step-4 transaminases is sufficient.

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7. Genes to Promote to Full `fetch-gene` Review

Priority for full individual review:

1. **PP_4481 (argD/astC, P59319, K00840)** — highest priority; missing from candidate metadata, dual catabolic/biosynthetic transaminase, needs formal step-4 assignment.
2. **PP_0372 (aruC, Q88QW2, K00821)** — promiscuous transaminase with primary bucket in ppu00300; confirm biosynthetic step-4 role vs. catabolic/lysine roles.

3. **PP_3571 (Q88GZ4, ArgE-family)** — resolve whether it is a genuine biosynthetic step-5 deacetylase or a promiscuous amidohydrolase.
4. **PP_1079 (argF) vs PP_1000 (arcB)** — document the K00611 anabolic/catabolic split so downstream curation does not confuse them.
5. **PP_0432 (argC1) vs PP_3633 (argC2)** — clarify whether both AGPR paralogs are biosynthetic or one is condition-specific.

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8. Mechanistic Model / Interpretation

The KT2440 pathway is best understood as a **redundant, promiscuity-tolerant biosynthetic network** overlaid on a strong catabolic arginine-handling capacity:

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               ┌─────────── DE NOVO BIOSYNTHESIS (module) ───────────┐
L-Glu ─ArgA/ArgJ─▶ NAG ─ArgB─▶ NAG-5-P ─ArgC1/ArgC2─▶ NAG-semialdehyde
                                                                 │
                                         AruC/ArgD (promiscuous) │ (4)
                                                                 ▼
L-Arg ◀─ArgH─ ArgSucc ◀─ArgG─ L-Cit ◀─ArgF─ L-Orn ◀─ArgE/ArgJ─ N-Ac-Orn
 (8)           (7)            (6, anabolic)   (5)
  ▲
  │  ═══════ CATABOLISM (out of module) ═══════
  └─ ArcA (deiminase) ▶ L-Cit ─ ArcB (catabolic OTCase) ▶ Orn + carbamoyl-P
                                ArcC (carbamate kinase) ▶ ATP
     AST pathway: shares AruC/ArgD transaminases with biosynthesis
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Two features distinguish KT2440 from the textbook *E. coli* model. (1) It uses the **cyclic ArgJ route** (acetyl recycling) in addition to linear ArgA/ArgE, which is metabolically economical because the acetyl group is retained rather than lost as acetate. (2) Step 4 lacks a dedicated ArgD; instead the **catabolic AST-pathway transaminases (AruC, ArgD/PP_4481) moonlight** to supply the biosynthetic transamination — a form of enzyme sharing between the anabolic and catabolic arms. This shared-enzyme architecture is why so many candidate genes (transaminases, OTCases) show paralog ambiguity: the biosynthetic and catabolic pathways use structurally similar enzymes and co-map in KEGG. Regulatory control (ArgR repression of *argF* and *carAB*) is the biological signal that cleanly separates the anabolic member (*argF*) from its catabolic paralog (*arcB*).

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9. Evidence Base

PMID	Title (abbrev.)	Organism	Supports	Transfer strength
22447897	ArgA/NAGS functional dissection	*P. aeruginosa*	Classical AAK-GNAT NAGS = PP_5185 step-1 assignment (F001)	Strong (close relative)
9393691	*aru* genes of AST pathway	*P. aeruginosa*	AruC / ArgD are AST N2-succinylornithine 5-aminotransferases; promiscuous step-4 (F002, F006)	Strong (close relative)
9286980	*argR* regulation	*P. aeruginosa* PAO1	*argF* = anabolic OTCase required for arg biosynthesis (F004)	Strong
1538697	*argF*/*aru* regulation	*P. aeruginosa* PAO	*argF* = arginine-repressible anabolic OTCase (F004)	Strong
15150254	ArgR controls *argF*	*P. syringae* pv. phaseolicola	*argF*/OCTase ArgR-repressed, generalizes across *Pseudomonas*	Moderate
1339419	*argJ* ornithine acetyltransferase	*N. gonorrhoeae*	ArgJ catalyzes acetylornithine→ornithine (step 5); complements *E. coli argE*/*argA*	Weak-moderate (distant, mechanism conserved)
23036925	Anoxic KT2440 strains	*P. putida* KT2440	Direct: anaerobic life induces arginine-fermentation (ADI) genes — confirms *arc* is catabolic	Direct (target)
23537069	Pressure transcriptome	*P. putida* KT2440	Direct: arginine deiminase pathway differentially expressed under stress	Direct (target)
23995642	D-amino acid metabolism	*P. putida* KT2440	Direct: KT2440 catabolizes D/L-arginine; racemase/catabolic context	Direct (target)

Direct target-organism evidence confirms KT2440 has an active catabolic arginine apparatus (ADI, AST, racemization) and is a prototroph. **The biosynthetic gene-to-step assignments rely primarily on strong homology transfer from *P. aeruginosa* PAO1** plus UniProt/KEGG annotation of the KT2440 proteome itself (verified for argA, argB, argH, argJ via UniProt REST, F005). No KT2440-specific enzymatic characterization of the individual biosynthetic Arg enzymes was located; this is the main evidence limitation.

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10. Limitations and Knowledge Gaps

1. **Homology-based step assignments.** Most biosynthetic gene→step calls rest on ortholog transfer from *P. aeruginosa* and on KEGG/UniProt automated annotation, not on direct KT2440 enzymology. Transfer is strong (same genus, high identity) but not experimentally verified in the target strain.
2. **Step 4 enzyme identity.** The precise in vivo contribution of AruC/PP_0372 vs ArgD/PP_4481 to biosynthetic (vs catabolic) transamination in KT2440 is unresolved. Both are promiscuous; kinetic partitioning is unknown.
3. **PP_3571 function.** Its role (biosynthetic step-5 deacetylase vs promiscuous amidohydrolase) is undetermined.
4. **ArgJ vs ArgA/ArgE flux.** Which acetyl-management route dominates physiologically is not established.
5. **Metadata completeness.** The candidate list omitted PP_4481, indicating the local metadata bucket did not capture all relevant transaminases — other cross-mapped genes may also be incompletely represented.

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11. Proposed Follow-up Experiments / Actions

Curation actions (immediate): - Add PP_4481 (argD/astC, P59319) to the ppu00220 module gene set (step 4). - Flag the ~20 out-of-boundary candidates as not_in_module (arcABC, carAB, ureABC, K01915 paralogs, gdhA/B, alaA, ansB). - Mark PP_3571 candidate_uncertain; promote PP_4481, PP_0372, PP_3571, argF/arcB to full fetch-gene review. - Add a curation note: step 4 served by promiscuous class-III AST transaminases (no dedicated ArgD).

Experimental (to resolve gaps): - **Gene knockouts:** single deletions of *argA*, *argB*, *argC1*/*argC2* (double), *argE*, *argF*, *argG*, *argH*, *argJ*, and *aruC*/*argD* (single and double), scoring arginine auxotrophy on minimal medium ± arginine. This would directly confirm which genes are essential for de novo biosynthesis in KT2440 — the single most decisive experiment. - **Enzyme assays:** recombinant AruC (PP_0372) and ArgD (PP_4481) tested for acetylornithine:2-oxoglutarate aminotransferase activity vs succinylornithine activity, to quantify biosynthetic vs catabolic partitioning. - **PP_3571 characterization:** acetylornithine deacetylase activity assay and knockout phenotype in an *argE* background. - **Regulatory confirmation:** verify ArgR-mediated arginine repression of PP_1079 (*argF*) but not PP_1000 (*arcB*) by qRT-PCR ± arginine, confirming the anabolic/catabolic split in KT2440 directly rather than by *P. aeruginosa* orthology.

Expert questions: - Does KT2440 ever require a dedicated ArgD, or is the promiscuous-transaminase solution complete under all growth conditions? - Is a GO-term request warranted to annotate "acetylornithine aminotransferase activity" onto the AST-type transaminases?

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12. Key References

- PMID: 22447897 — Functional dissection of ArgA/NAGS in *P. aeruginosa*.
- PMID: 9393691 — Cloning of the *aru* genes (AST catabolic pathway) in *P. aeruginosa*.
- PMID: 9286980 — *argR* regulation of arginine biosynthesis/catabolism in *P. aeruginosa* PAO1.
- PMID: 1538697 — Regulation of anabolic *argF* and catabolic *aru* genes.
- PMID: 15150254 — ArgR control of *argF* in *P. syringae*.
- PMID: 1339419 — *argJ* ornithine acetyltransferase (*N. gonorrhoeae*), complements *E. coli* *argE*/*argA*.
- PMID: 23036925 — Anoxic KT2440 induces arginine-fermentation/ADI genes.
- PMID: 23537069 — KT2440 transcriptome: ADI pathway stress-sensitive.
- PMID: 23995642 — D-amino acid metabolism in KT2440 (arginine catabolism/prototrophy context).

Consensus

The acetylated-ornithine L-arginine biosynthesis module is complete and satisfiable in *P. putida* KT2440: all eight steps are encoded, with redundant implementations at the acetyl-management steps (ArgA + bifunctional ArgJ at step 1; hydrolytic ArgE + ArgJ transacetylase at step 5) and two ArgC paralogs at step 3. Step 4 uses no dedicated ArgD but promiscuous AST-type transaminases — *aruC*/PP_0372 and *argD*/PP_4481 (the latter missing from the candidate list and needing to be added). Only ~10–11 of the 31 candidates are true pathway members; urease, glutamine-synthetase/polyamine paralogs, glutamate dehydrogenases, the catabolic *arc* (arginine-deiminase) operon, and carbamoyl-phosphate synthase are out-of-boundary over-propagations, and the biosynthetic OTCase is *argF*/PP_1079, not its catabolic K00611 paralog *arcB*/PP_1000.
